

Problem 15.10

Find the present value of an annuity which pays \$200 at the end of each quarter-year for 12 years if the rate of interest is 6% convertible quarterly.

Solution

The nominal interest rate is 6% convertible quarterly, so the effective interest rate per quarter is

$$i = \frac{0.06}{4} = 0.015.$$

Since payments are made at the end of each quarter-year for 12 years, the number of payments is

$$n = 12(4) = 48.$$

The present value is the value of an annuity-immediate with quarterly payments of 200:

$$PV = 200a_{\overline{48}|}.$$

Using the present value formula,

$$a_{\overline{48}|} = \frac{1 - (1.015)^{-48}}{0.015}.$$

Therefore,

$$PV = 200 \left(\frac{1 - (1.015)^{-48}}{0.015} \right).$$

Finally,

$$PV \approx 6808.510729.$$

Hence, the present value of the annuity is

$$\boxed{\$6,808.51}.$$